

Magnetite graphene oxide (MGO) incorporated hollow fiber membrane for pharmaceutical wastewater treatment

Harshita Dihora¹, Mrinmoy Mondal¹, Puyam S. Singh¹

¹Membrane Science and Separation Technology Division, CSIR-CSMCRI, G.B. Marg, Bhavnagar 364021, India.

E-mail address: harshitadihora@gmail.com

Introduction

- Pharmaceutical Pollution:** The presence of persistent Pharmaceutical Active Compounds (PhACs) in industrial wastewater poses significant ecological and health risk [1].
- Treatment Limitations:** Conventional biological and chemical processes often fail to remove recalcitrant PhACs, leading to high reagent costs, significant sludge production, and the formation of toxic by-products.
- Need for Advanced Treatment:** The increasing discharge of pharmaceutical wastewater requires the development of efficient, sustainable, and high-performance separation technologies to effectively remove contaminants.
- Sustainable Membrane Technology:** Hollow fiber membranes offer a high-efficiency, energy-effective alternative for industrial-scale purification [2].
- MGO enhancement:** Incorporating MGO improves membrane performance by increasing surface area, hydrophilicity, and the availability of functional active sites, thereby enhancing permeability and contaminant rejection.

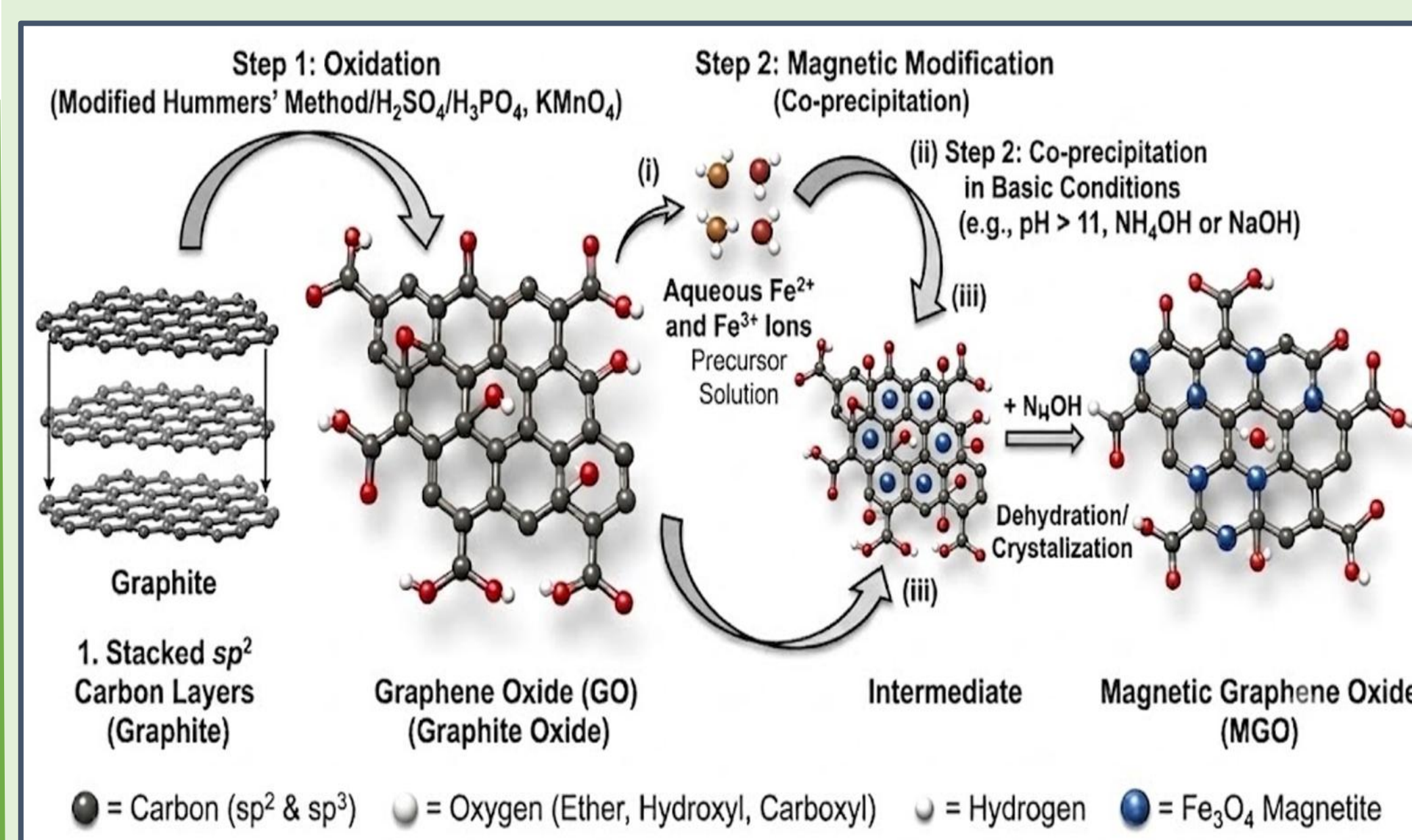


Fig. 1: Schematic illustration of GO synthesis from graphite via the modified Hummers' method and subsequent (Fe_3O_4) incorporation through co-precipitation to form MGO.

GO and MGO synthesis process

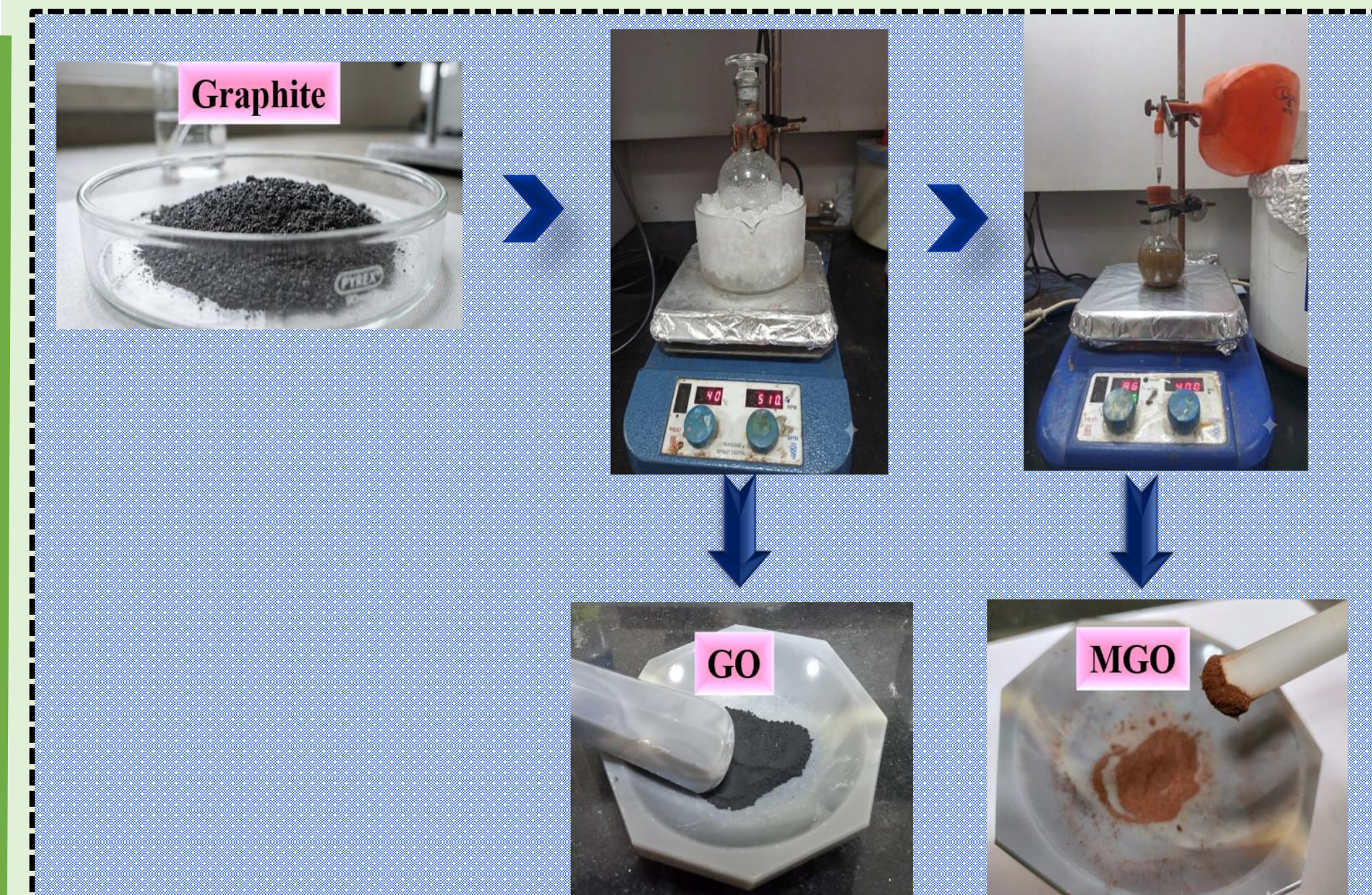


Fig. 2: Optical images illustrating the synthesis pathway from graphite to GO and final MGO nanocomposites.

Material Charecterization

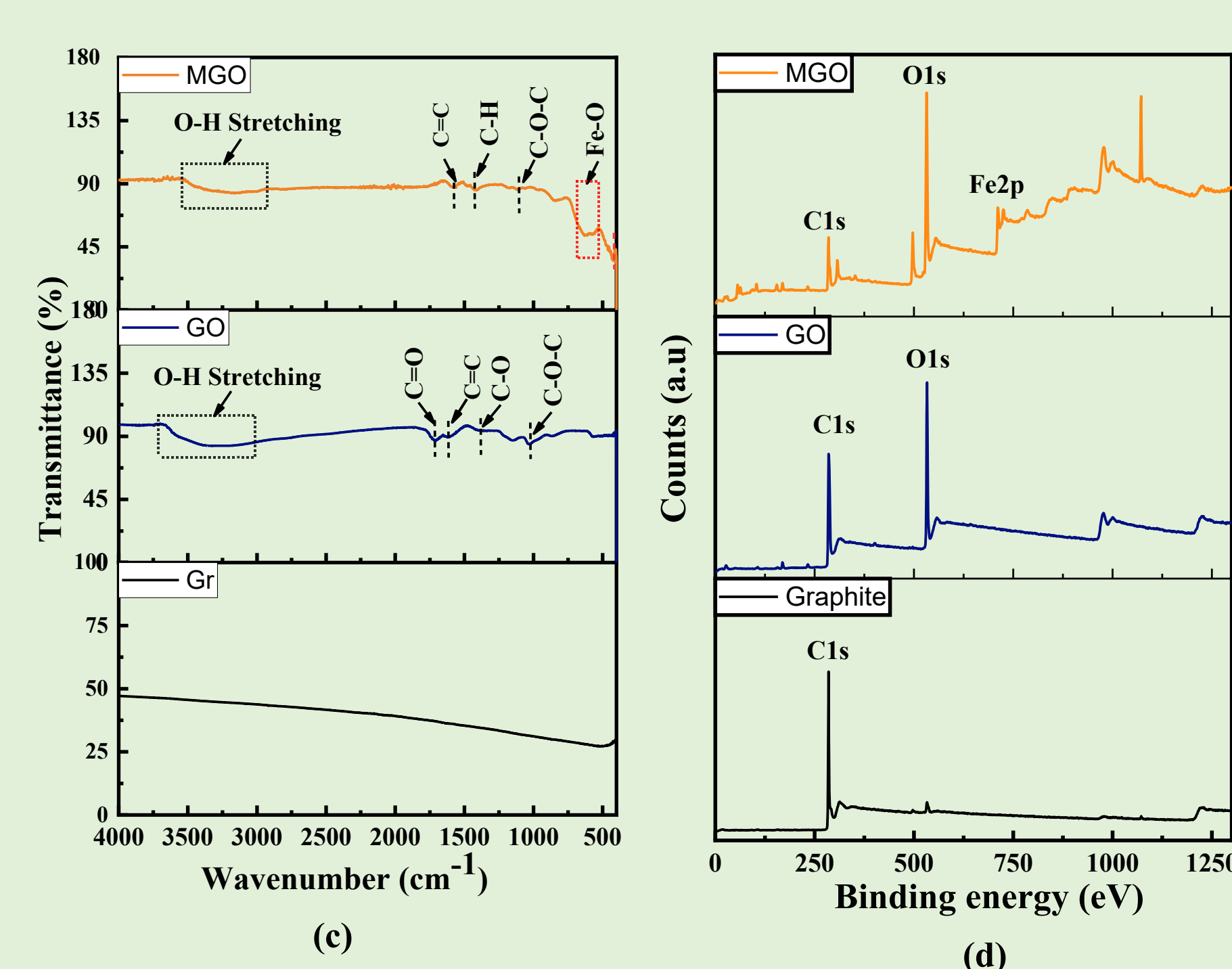
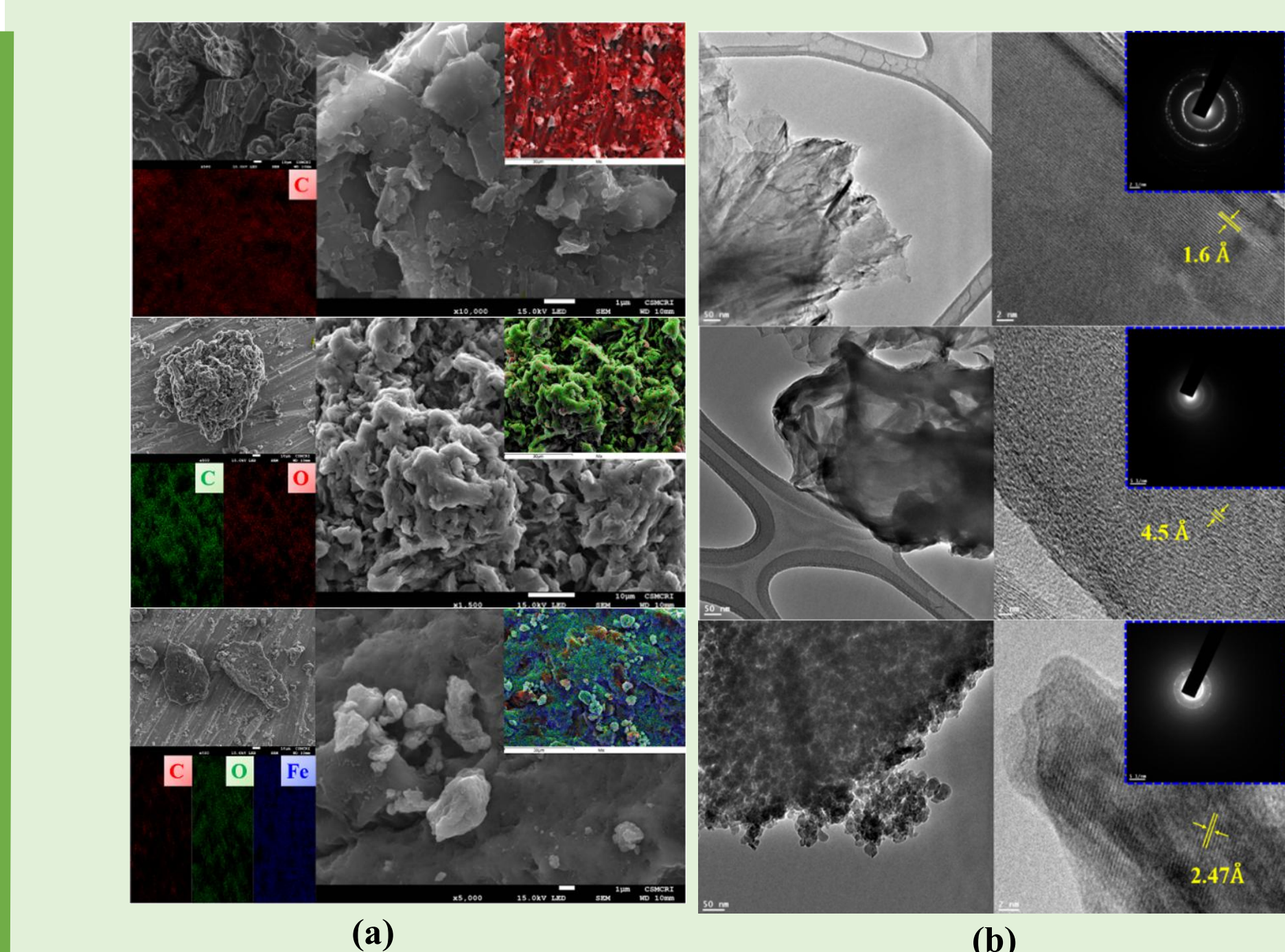


Fig. 3: (a) FE-SEM images with colour mapping, (b) HR-TEM images with fringes and SAED, (c) ATR-IR spectra, (d) XPS analysis survey analysis of Graphite, GO and MGO.

Hollow fiber membrane preparation

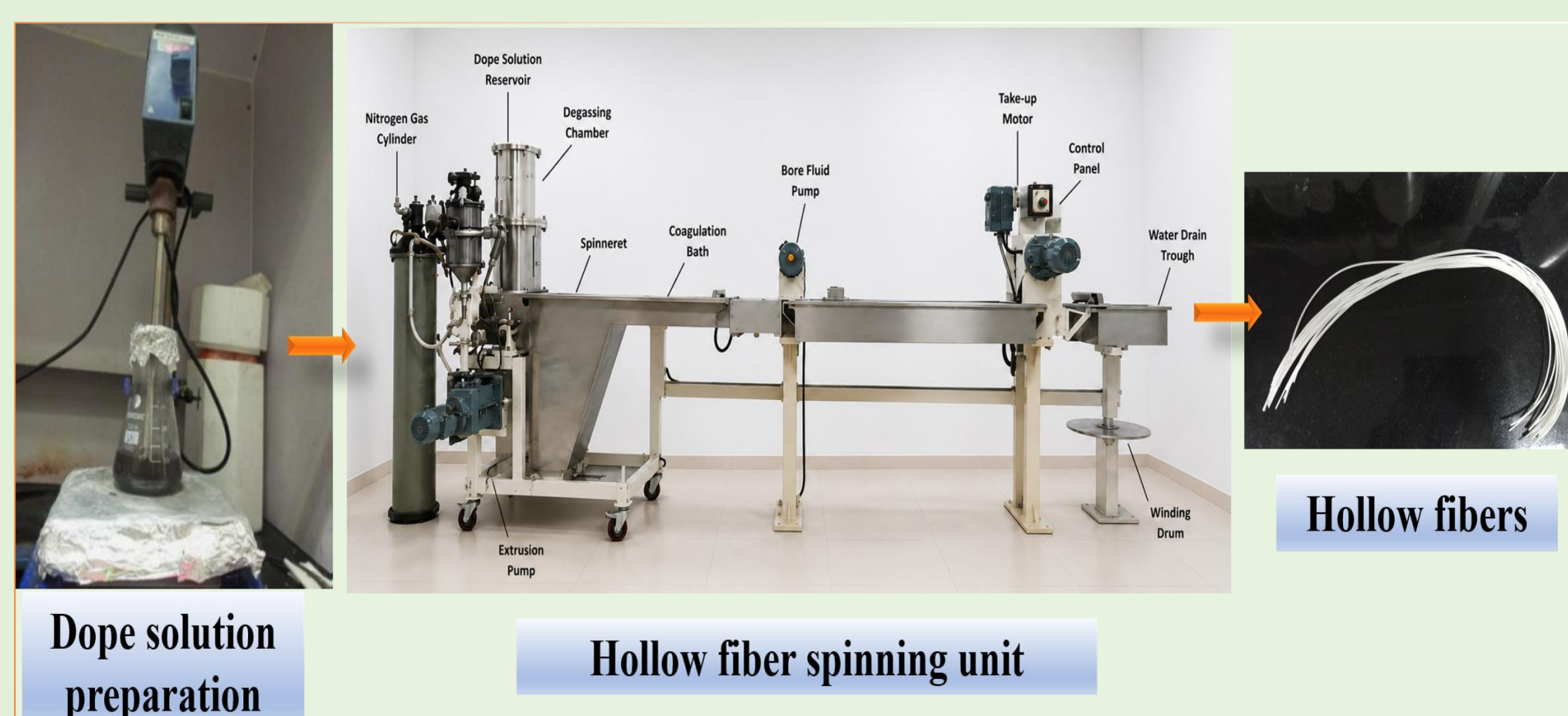


Fig. 4: Schematic representation of the hollow fiber membrane fabrication process, spanning from initial dope solution preparation to the final spinning and collection of hollow fibers.

Cross-flow membrane filtration set-up

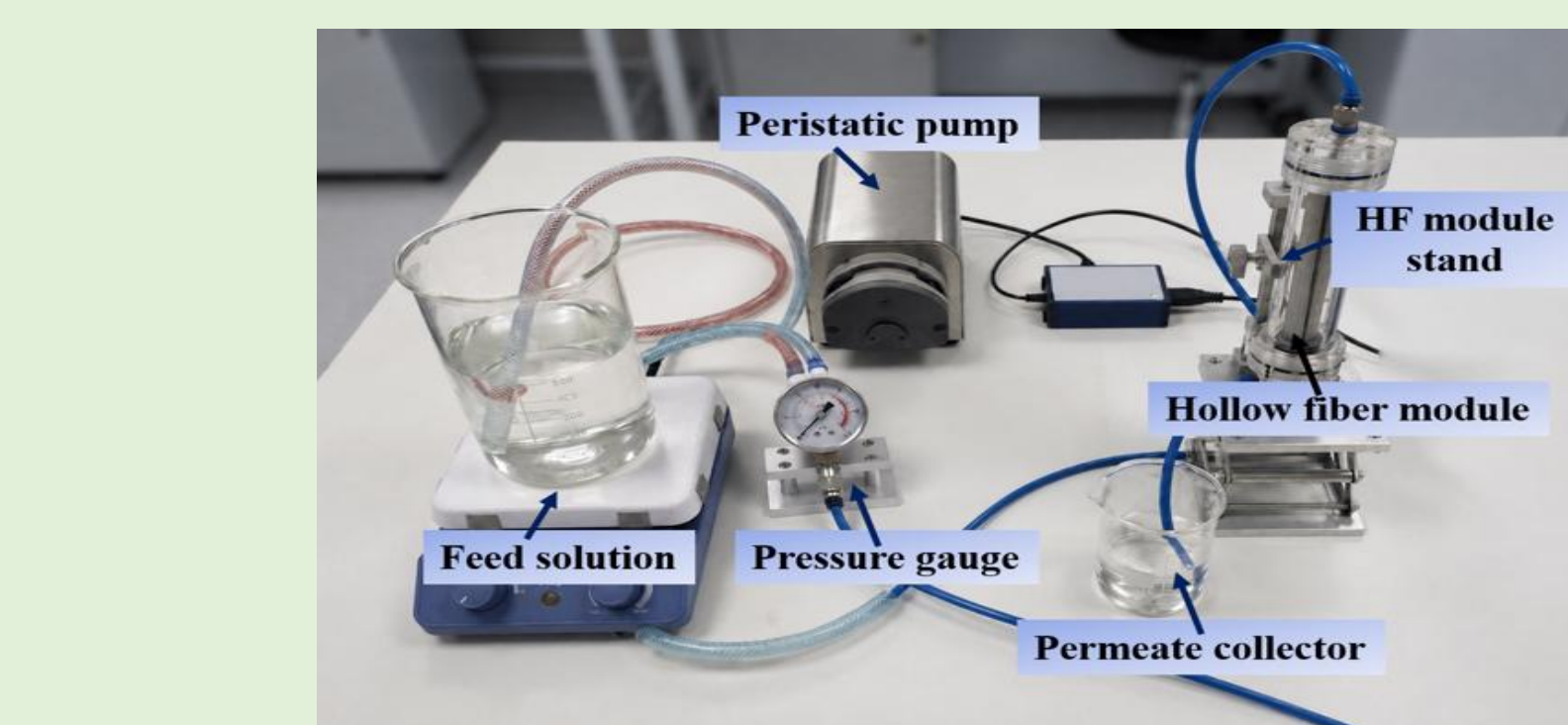


Fig. 5: Optical image showing the experimental configuration of the crossflow hollow fiber filtration system used for membrane performance testing.

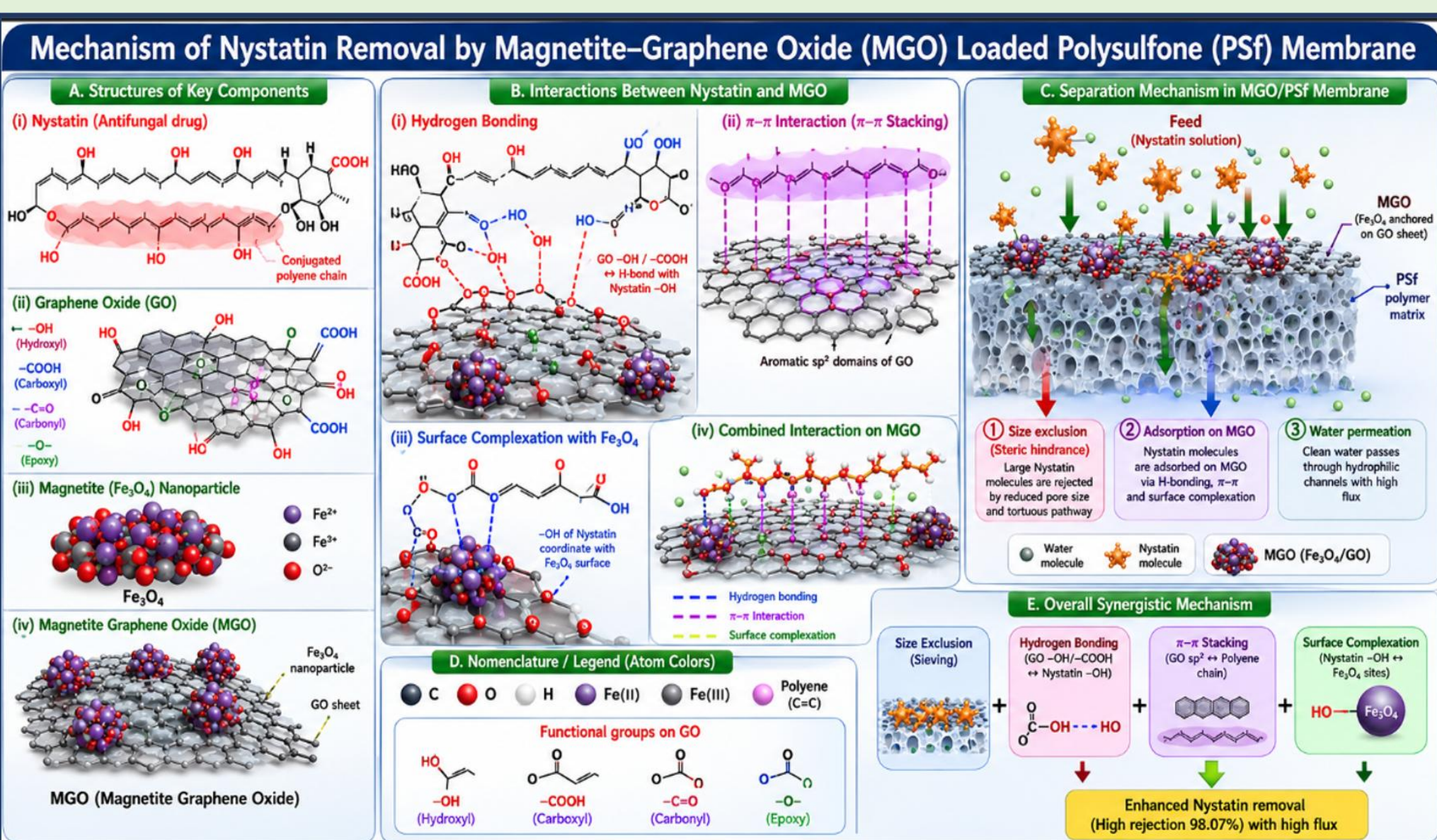
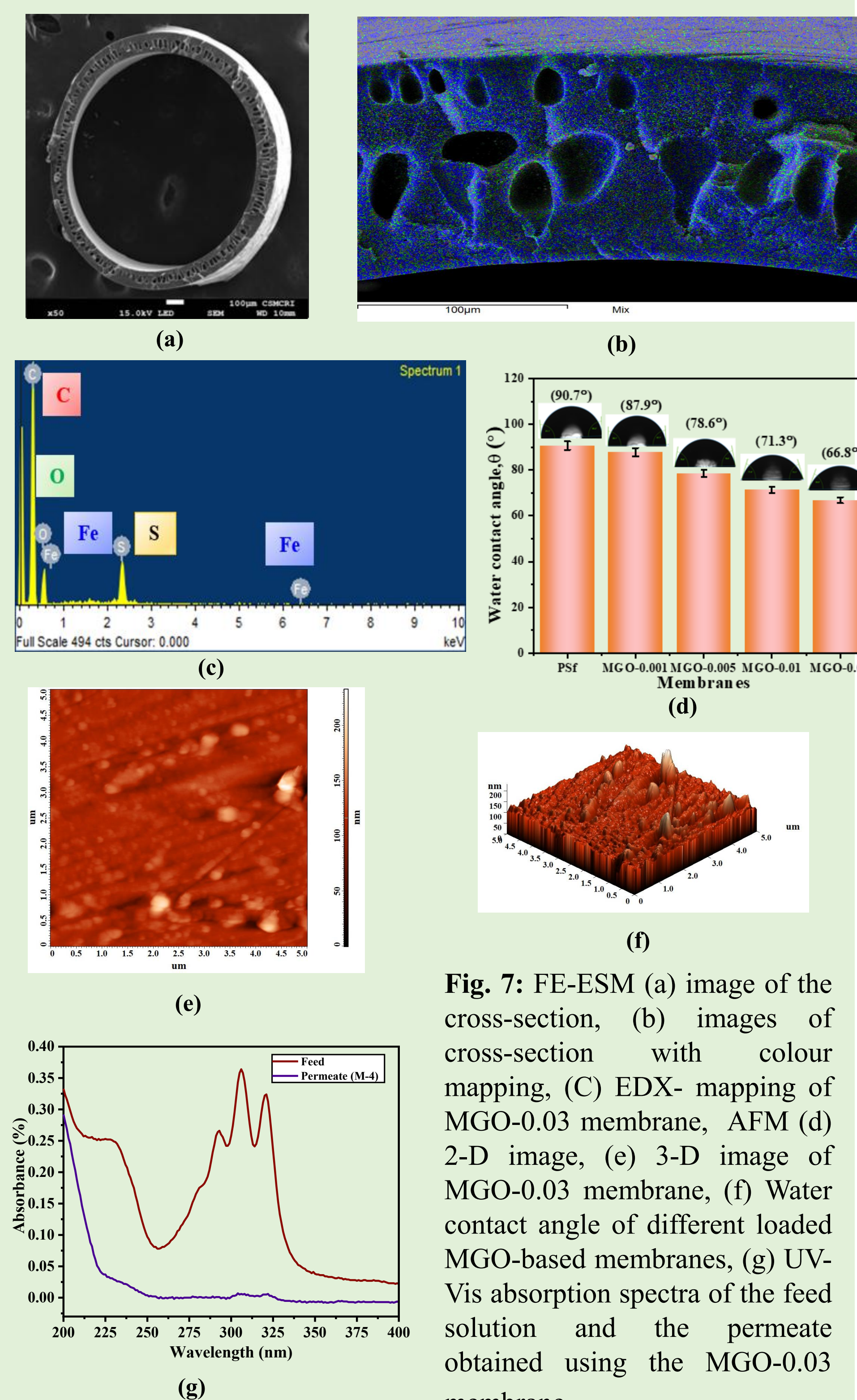


Fig.6: Proposed multi-faceted mechanism for nystatin removal by MGO-loaded PSf membranes, encompassing size exclusion, adsorption, and surface complexation. pathways.

Membrane Charecterization



Performance

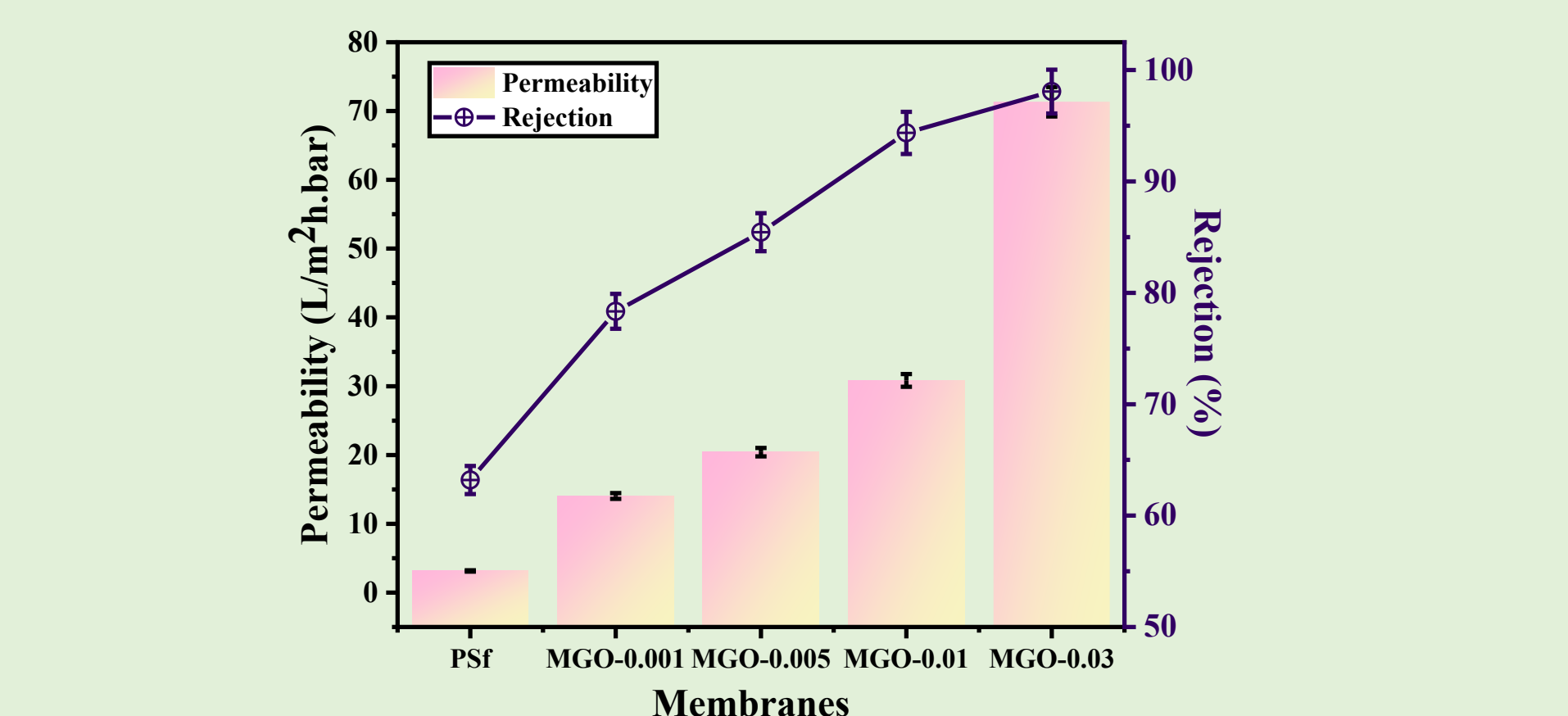


Fig.8: Permeate flux and nystatin rejection of membranes.

Result

- Optimal Performance:** 0.03% MGO membrane showed high nystatin rejection and water permeability.
- Enhanced Properties:** MGO improved membrane hydrophilicity and morphology.
- Separation Mechanism:** Nystatin was removed via size-exclusion and surface interactions.
- Application:** Promising membrane for efficient pharmaceutical wastewater treatment.

Reference

- [1] Y. Luo, W. Guo, H.H. Ngo, L.D. Nghiem, F.I. Hai, J. Zhang, S. Liang, X.C. Wang, A review on the occurrence of micropollutants in the aquatic environment and their fate and removal during wastewater treatment, Science of The Total Environment 473–474 (2014) 619–641. <http://doi.org/10.1016/j.scitotenv.2013.12.065>.
- [2] B. Van der Bruggen, L. Lejon, C. Vandecasteele, Reuse, Treatment, and Discharge of the Concentrate of Pressure-Driven Membrane Processes, Environ. Sci. Technol. 37 (2003) 3733–3738. <https://doi.org/10.1021/es0201754>.

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